ANALYSTLAB AFRICA

EXPERIENCE LAB INTERNSHIP PROGRAMME

WORK BREAKDOWN STRUCTURE

Consumer Complaint Trends and Response Intelligence Platform

Week 4 • Batch F

Track: Project Management

Status: Proposed planning document

1. Document Purpose

This Work Breakdown Structure (WBS) decomposes the total scope of the Consumer Complaint Trends and Response Intelligence Platform into manageable project components and work packages. It provides the scope-based structure that will support detailed scheduling, responsibility assignment, resource planning, progress monitoring and project control.

The WBS reflects the five internship tracks identified in the Week 4 Batch F assignment: Project Management, Data Analytics, Data Science, Machine Learning and Generative AI. It also includes shared data-readiness, integration and final delivery activities because the assignment requires the five tracks to contribute to one integrated project.

2. WBS Structure and Numbering

The WBS uses a hierarchical numbering convention. Level 1 represents the overall project; Level 2 represents major project components or workstreams; Level 3 identifies sub-components; and Level 4 identifies work packages or specific activities where useful.

WBS level	Description	Illustration
Level 1	Overall project	1.0 Consumer Complaint Intelligence Platform
Level 2	Major project component / workstream	2.0 Data Analytics
Level 3	Sub-component	2.1 Data inspection and preparation
Level 4	Work package / activity	2.1.1 Inspect dataset structure and field types

The WBS identifies the work required; it does not assign calendar dates or durations. Those will be developed in the separate project schedule (Deliverable 3). Individual task owners and detailed accountability will be assigned through the RACI matrix (Deliverable 4).

3. Work Breakdown Structure

WBS ID	WBS component / work package	Description / output
1.0	Consumer Complaint Trends and Response Intelligence Platform	Complete integrated project scope.
1.1	Project Management	Plan, coordinate, monitor and control the integrated project.
1.1.1	Project initiation	Establish project purpose, objectives, scope and governance.
1.1.1.1	Develop project charter	Document objectives, scope, outcomes, stakeholders, deliverables, assumptions and dependencies.
1.1.1.2	Identify stakeholders	Identify programme, mentor, project and workstream stakeholders.
1.1.1.3	Define deliverables and acceptance criteria	Specify expected outputs and evidence required for acceptance.
1.1.2	Project planning	Develop the integrated baseline and coordination arrangements.
1.1.2.1	Develop WBS	Decompose the project scope into work packages.
1.1.2.2	Develop project schedule	Sequence activities and identify proposed durations, owners and milestones.
1.1.2.3	Prepare RACI matrix	Allocate Responsible, Accountable, Consulted and Informed roles.
1.1.2.4	Prepare communication plan	Define stakeholder information needs, channels, frequency and owners.
1.1.3	Risk, issue and change management	Identify and control threats, issues and proposed changes.
1.1.3.1	Identify and assess project risks	Assess probability, impact and priority of project risks.
1.1.3.2	Maintain issue register	Record actual blockers and issues, owners, actions and status.
1.1.3.3	Assess and control changes	Document change requests and assess scope, schedule, resource and quality impacts.
1.1.4	Project monitoring and coordination	Monitor workstream delivery and cross-track coordination.
1.1.4.1	Maintain task and milestone tracker	Record owners, planned dates, progress, blockers and status.
1.1.4.2	Monitor schedule and dependencies	Compare progress with the plan and follow up on cross-track hand-offs.
1.1.4.3	Coordinate peer reviews	Arrange and record review of relevant outputs.
1.1.4.4	Track acceptance criteria	Monitor evidence supporting completion and acceptance.
1.1.4.5	Prepare progress reports	Report progress, risks, issues, decisions and next steps.
1.1.5	Project closure and handover	Coordinate final review, reporting and transfer of completed outputs.
1.1.5.1	Coordinate integration review	Check consistency and readiness of outputs across workstreams.
1.1.5.2	Compile final project status report	Summarise delivery status, outstanding items and project position.
1.1.5.3	Compile handover documentation	Organise final deliverables, supporting documentation and evidence.
1.1.5.4	Prepare final presentation	Explain work, results, limitations and individual contribution.
2.0	Shared Data Readiness and Governance	Provide consistent, validated data foundations for the technical tracks.
2.1	Dataset inspection	Understand the supplied source data.
2.1.1	Review supplied files and data dictionary	Confirm available files, fields, definitions and reporting period.
2.1.2	Inspect dataset structure and field types	Review worksheet, record count, columns and data types.
2.1.3	Check Complaint ID uniqueness and duplicates	Investigate candidate primary-key uniqueness and duplicate records.

2.2	Data quality assessment and preparation	Identify and document data-quality matters.
2.2.1	Investigate missing values	Assess missingness in sub-issue, public response and other fields.
2.2.2	Validate date fields	Check submitted and received dates and identify unusual or inconsistent values.
2.2.3	Review categorical consistency	Inspect product, issue, response and other categorical values.
2.2.4	Document cleaning decisions	Record transformations, exclusions, standardisation and rationale.
2.2.5	Preserve original dataset	Retain an unchanged source copy and maintain reproducible preparation steps.
2.2.6	Validate prepared data and calculations	Reconcile record counts and test derived fields and calculations.
2.3	Shared definitions and data controls	Establish consistent interpretation across workstreams.
2.3.1	Define complaint volume and reporting periods	Document count basis, date basis, filters and period treatment.
2.3.2	Define product and issue classifications	Document use of product, sub-product, issue and sub-issue.
2.3.3	Define company response classifications	Document the meaning and treatment of response fields.
2.3.4	Define timely-response calculation	Use valid Yes/No records in the denominator and report excluded records separately.
2.3.5	Define CFPB forwarding interval	Calculate Date received less Date submitted; do not describe it as company resolution time.
3.0	Data Analytics Workstream	Develop complaint monitoring and response dashboard.
3.1	KPI and analytical preparation	Prepare measures required for dashboard reporting.
3.1.1	Calculate complaint volume measures	Calculate total, annual and monthly complaint volumes.
3.1.2	Calculate product and issue measures	Calculate complaints by product, sub-product, issue and sub-issue.
3.1.3	Calculate response measures	Calculate response-category distribution, timely-response percentage and in-progress count.
3.1.4	Calculate forwarding interval measures	Calculate average and median forwarding interval and related comparisons.
3.1.5	Calculate year-over-year changes	Calculate and document annual complaint-volume changes.
3.2	Dashboard development	Build the required interactive dashboard pages.
3.2.1	Develop executive complaint overview	Present total complaints, trends, product distribution and key response/interval KPIs.
3.2.2	Develop product and issue analysis	Present product, sub-product, issue and product-versus-issue analysis.
3.2.3	Develop company response analysis	Present response categories, timely/untimely distribution and in-progress complaints.
3.2.4	Develop complaint timeline analysis	Present monthly and annual trends and forwarding interval analysis.
3.2.5	Configure filters and visual design	Add relevant filters, clear titles, KPI definitions and suitable visuals.
3.3	Dashboard validation and insights	Validate outputs and prepare evidence-based interpretation.
3.3.1	Validate dashboard calculations	Reconcile visuals and KPI results to validated calculations.
3.3.2	Develop business insights	Prepare at least five evidence-based findings.
3.3.3	Develop recommendations	Prepare at least three actionable recommendations supported by findings.
3.3.4	Prepare analytics documentation	Provide cleaning process, scripts, KPI definitions, README and data-quality documentation.
4.0	Data Science Workstream	Investigate complaint patterns and response trends.
4.1	Exploratory data analysis	Describe distributions, trends and patterns.
4.1.1	Analyse annual and monthly distributions	Examine complaint volumes over time.
4.1.2	Analyse product and issue distributions	Identify frequently reported products and issues.
4.1.3	Analyse response categories	Examine recorded company response distributions.
4.1.4	Analyse forwarding intervals	Examine interval distributions and differences across categories.
4.2	Research and statistical analysis	Formulate and test suitable research questions.
4.2.1	Develop research questions and hypotheses	Document at least three questions or hypotheses.
4.2.2	Select appropriate statistical methods	Assess assumptions and select suitable descriptive or inferential methods.
4.2.3	Conduct statistical analysis	Apply appropriate tests, such as chi-square or non-parametric tests where justified.
4.2.4	Interpret results and limitations	Explain results without confusing association with causation.
4.3	Analysis reporting and reproducibility	Document methods and supported findings.
4.3.1	Prepare reproducible analysis notebook	Record data preparation, analysis and visualisations.
4.3.2	Document at least three supported findings	Link findings to results and evidence.
4.3.3	Prepare analytical report	Document interpretation, limitations and alternative explanations.
5.0	Machine Learning Workstream	Develop and evaluate a complaint-volume forecasting prototype.
5.1	Prediction problem and modelling data	Define the target and prepare the time series.
5.1.1	Define monthly complaint-volume target	Specify target measure and aggregation basis.
5.1.2	Aggregate complaints by month	Create monthly complaint counts.
5.1.3	Investigate missing reporting periods	Distinguish missing months from genuine zero-complaint months.
5.1.4	Assess partial first and last months	Determine appropriate treatment of May 2017 and August 2023 partial periods.
5.1.5	Prepare time-based features and data splits	Create features and chronological training, validation and testing periods.
5.2	Forecast model development	Develop baseline and comparison methods.
5.2.1	Establish baseline forecast	Create a baseline forecasting approach.
5.2.2	Develop comparison models	Compare at least two suitable methods where feasible.
5.2.3	Apply chronological validation	Use chronological testing or rolling-origin validation and prevent future-data leakage.
5.3	Model evaluation and reporting	Evaluate performance and explain limitations.
5.3.1	Calculate evaluation metrics	Use suitable metrics such as MAE, RMSE or WAPE.
5.3.2	Compare forecasts with actual volumes	Visualise forecast and actual values.
5.3.3	Conduct error analysis	Examine prediction errors and their implications.
5.3.4	Document model limitations and recommendations	Explain practical use, limitations and appropriate interpretation.
5.3.5	Prepare reproducible modelling workflow	Provide notebook/source code and saved model where appropriate.
6.0	Generative AI Workstream	Develop a data-grounded consumer complaint insights assistant.

6.1	Knowledge and requirements preparation	Define supported questions and validated sources.
6.1.1	Review dataset and data dictionary	Identify available fields, meanings and limitations.
6.1.2	Identify answerable and unsupported questions	Define question types the data can and cannot support.
6.1.3	Document KPI definitions and data sources	Use agreed definitions and validated analytical outputs.
6.2	Assistant design and development	Build the prototype and grounding approach.
6.2.1	Define system instructions and safeguards	Set boundaries against invented allegations and legal conclusions.
6.2.2	Configure knowledge or analytical access	Use controlled analytical functions or retrieval where practical.
6.2.3	Develop AI assistant prototype	Implement the question-and-answer experience.
6.3	Evaluation and documentation	Test response quality and document limitations.
6.3.1	Prepare evaluation questions	Include at least 20 valid, ambiguous and unsupported questions.
6.3.2	Test numerical grounding and source use	Verify numerical answers against actual calculations and sources.
6.3.3	Evaluate accuracy and consistency	Record evaluation results and areas for improvement.
6.3.4	Document limitations and safeguards	Provide README, configuration and responsible-use documentation.
7.0	Integration, Quality Review and Final Delivery	Bring completed workstream outputs together and prepare final delivery.
7.1	Cross-track integration	Coordinate the shared outputs and hand-offs.
7.1.1	Confirm consistent metric definitions	Check consistency of complaint volume, products, issues, responses, timeliness and forwarding interval.
7.1.2	Review dashboard and statistical findings alignment	Check that shared figures and interpretations are consistent.
7.1.3	Review forecasting output and limitations	Confirm forecast scope, evaluation and limitations are communicated.
7.1.4	Review AI assistant grounding	Confirm responses use validated information and appropriate caveats.
7.2	Quality and acceptance review	Check evidence against requirements.
7.2.1	Conduct cross-track peer review	Record review comments and required revisions.
7.2.2	Verify deliverable acceptance criteria	Check outputs against agreed criteria and retain evidence.
7.2.3	Track and resolve review findings	Record defects, actions, owners and closure status.
7.3	Final submission and handover	Organise final project evidence and communicate completion.
7.3.1	Compile final project documentation	Organise README files, methods, assumptions and limitations.
7.3.2	Compile individual contribution evidence	Record actual contributions and meaningful revisions after feedback.
7.3.3	Prepare final project presentation	Explain outputs, results, limitations and contribution to the integrated project.
7.3.4	Complete final handover	Provide completed and tested outputs with supporting documentation.

4. Work Package Guidance

The following work-package guidance explains how selected WBS components should be controlled. Detailed dates and named owners will be assigned in the project schedule and RACI matrix.

WBS area	Key input	Expected output / completion evidence
1.1 Project Management	Assignment requirements, charter and workstream updates	Integrated plan, registers, progress records, review evidence and handover.
2.0 Shared Data Readiness	Supplied Excel dataset and data dictionary	Documented data-quality assessment, reproducible preparation and shared metric definitions.
3.0 Data Analytics	Validated dataset and agreed KPI definitions	Four-page interactive dashboard, KPI calculations, five or more findings and three or more recommendations.
4.0 Data Science	Prepared dataset and shared definitions	Reproducible EDA/statistical analysis, at least three research questions or hypotheses and at least three supported findings.
5.0 Machine Learning	Monthly complaint series and documented period treatment	Baseline and model comparison, chronological validation, evaluation metrics, forecasts and error analysis.
6.0 Generative AI	Validated dataset, metric definitions and analytical outputs	Working assistant prototype, at least 20 evaluation questions, evaluation results and safeguards.
7.0 Integration and Final Delivery	Completed workstream outputs and review evidence	Consistent integrated outputs, documented review findings, final report, presentation and handover.

5. WBS Control Notes and Boundaries

- The WBS covers the integrated project described in the Week 4 assignment; it does not mean that the Project Management intern personally performs every technical activity.
- Each technical workstream remains responsible for its own technical deliverables. The Project Manager coordinates dependencies, monitoring, reviews and integration.
- The WBS is scope-oriented. Dates, durations, sequence, resource loading and progress status belong in the project schedule and task tracker.

- The WBS includes the assignment's stated technical requirements so that the Project Manager can coordinate the complete integrated project.
- The WBS does not assume that any technical output, meeting, approval, test or handover has already occurred.
- If an activity is added or removed, the change should be assessed against the charter and managed through the agreed change-control process.

6. Source and Alignment

This WBS is based on the AnalystLab Africa Experience Lab Internship Programme, Official Weekly Assignment, Week 4: Consumer Complaint Trends and Response Intelligence Platform, Version 1.0. It reflects the Project Management requirements on pages 10–12, the shared integration requirements on page 12, and the technical track activities and deliverables on pages 4–10.